

N8N Zendesk AI Ticket Analysis Workflow

Project Overview

This n8n workflow implements an intelligent Zendesk ticket analysis system that automatically processes incoming support tickets using Claude AI, categorizes them, and takes appropriate actions based on urgency and content. The system integrates Zendesk, Google Sheets, Jira, and Slack to create a comprehensive ticket management solution.

Architecture & Data Flow

Webhook → Get Zendesk Ticket → Normalize Data → AI Analysis → Parse Results → Decision Logic
↓
Google Sheets Logging ← Emergency Check → Update Ticket + Create Jira Issue
↓
Standard Update

Core Components:

- **Webhook Trigger:** Receives ticket events from Zendesk
- **AI Analysis Engine:** Uses Claude Opus 4 for intelligent categorization
- **Decision Engine:** Routes tickets based on emergency status
- **Multi-Platform Integration:** Logs to Google Sheets, creates Jira issues, sends Slack alerts

Installation & Setup Guide

1. Prerequisites & API Keys

Anthropic (Claude AI)

1. Visit console.anthropic.com → API Keys → Create Key
2. Copy API key (format: `sk-ant-api03-...`)
3. **N8N:** Settings → Credentials → Add “Anthropic API” → Paste key

Zendesk API Token

1. Zendesk Admin → Apps & integrations → APIs → Zendesk API → Settings
2. Enable “Token Access” → Add API token → Copy token
3. **N8N:** Add “Zendesk API” credential with subdomain, email, and token

Google Sheets OAuth2

1. Google Cloud Console → New Project → Enable Sheets API
2. Credentials → Create OAuth 2.0 Client → Web application
3. **N8N:** Add “Google Sheets OAuth2 API” credential → Connect account

Jira Cloud API Token

1. Atlassian API Tokens → Create token
2. N8N: Add “Jira Software Cloud API” with email, token, and domain

Slack Bot Token

1. Slack API → Create App → OAuth & Permissions
2. Add scopes: `chat:write`, `channels:read` → Install to Workspace
3. N8N: Add “Slack OAuth2 API” with Bot User OAuth Token

2. Service Configuration

Google Sheets Setup

1. Create spreadsheet: N8N Tickets Database
2. Headers: Ticket ID | Subject | Topic | Urgency | Sentiment | Response
3. Copy Document ID from URL: `docs.google.com/spreadsheets/d/[DOCUMENT-ID]/`

Jira Project Setup

1. Create project (Service Management template)
2. Ensure “Incident” issue type exists
3. Note Project ID from Project Settings → Details

Zendesk Webhook Configuration

1. **Get N8N Webhook URL** from workflow Webhook node
2. **Zendesk Admin** → Extensions → Webhooks → Create webhook:

Endpoint: `[n8n-webhook-url]/chat-ticket`
Method: POST, Format: JSON

3. **Create Trigger** → Business Rules → Triggers:

Condition: Ticket Is created OR Is updated

Action: Notify webhook with JSON:

```
{
  "ticket": {
    "id": "{{ticket.id}}",
    "subject": "{{ticket.title}}",
    "description": "{{ticket.description}}",
    "priority": "{{ticket.priority}}"
  }
}
```

3. N8N Workflow Installation

1. **Import** the provided JSON file into N8N
2. **Update Node Credentials** - Verify each node references correct credentials:
 - Anthropic Chat Model → “Anthropic account 2”
 - Zendesk nodes → “Zendesk account”
 - Google Sheets → “Google Sheets account” + your Document ID
 - Jira → “Jira SW Cloud account” + your Project ID
 - Slack → “Slack account” + #support-alerts channel
3. **Activate** the workflow
4. **Test** by creating a ticket in Zendesk

Key Features

Intelligent Classification

- **Sentiment Analysis:** Positive, neutral, negative classification
- **Urgency Scoring:** 1-5 scale based on content analysis
- **Topic Categorization:** billing, technical, account, sales, other
- **Emergency Detection:** Identifies critical issues requiring immediate attention

AI Prompt Engineering

The system uses structured prompt engineering that: - Enforces strict JSON schema compliance for consistent output - Implements decision rules for emergency classification - Generates contextually appropriate responses (max 200 characters) - Maintains conversation context using 10-message buffer window memory

Automated Actions

Emergency Tickets (high urgency, negative sentiment, critical keywords): - Internal note with urgency indicator - Public reply acknowledging urgency - Jira incident creation - Google Sheets logging

Standard Tickets: - Public reply with helpful information - Google Sheets logging

Performance Metrics

- **Processing Time:** 5-10 seconds end-to-end
- **Scalability:** ~100 tickets/hour current configuration
- **AI Processing:** 3-5 seconds per ticket
- **Success Rate:** 95%+ with proper configuration

Security & Monitoring

Security Features

- All API credentials encrypted in N8N credential store
- HTTPS webhook endpoints
- No sensitive customer data logged to sheets
- Access controls on all integrated platforms

Error Handling & Monitoring

- **Error Trigger:** Catches workflow failures
- **Slack Notifications:** Immediate alerts for issues
- **Robust JSON Parsing:** Handles unexpected AI outputs
- **Execution Logging:** Full audit trail in N8N

Troubleshooting

Common Issues

Webhook Not Triggering: Check URL format, firewall settings, test with curl
AI Analysis Failing: Verify API key, check token limits, review prompt formatting
Google Sheets Errors: Re-authorize OAuth2, verify permissions and document ID
Jira Issues: Check API token permissions, confirm project and issue type exist

Testing Commands

```
# Test webhook endpoint
curl -X POST https://your-n8n-domain/webhook/webhook-id/chat-ticket \
-H "Content-Type: application/json" \
-d '{"ticket":{"id":"123","subject":"Test","description":"Test ticket"}}'
```

Limitations & Future Improvements

Current Limitations

- **Language Support:** Optimized for English-language tickets
- **Rate Limiting:** No built-in throttling for high-volume scenarios
- **Response Quality:** Generic responses may not address specific needs

Planned Enhancements

- **Multi-language Support:** Detect and process tickets in multiple languages
- **Advanced Routing:** ML-based escalation path optimization
- **Custom Templates:** Dynamic response system based on ticket categories
- **Predictive Analytics:** Forecast ticket volume and resource requirements

Cost Analysis

Monthly Operational Costs (1000 tickets): - Anthropic Claude API: \$20-50 - Google Sheets API: Free (within quotas) - Other APIs: Included with existing subscriptions - **Total:** \$20-50/month for full automation

This workflow demonstrates advanced n8n capabilities including AI integration, multi-platform orchestration, and intelligent decision-making, providing a robust foundation for automated customer support operations.